Inter-AS Routing (BGP)

1 The Role of BGP: The "Internet's Glue"

While **Intra-AS** protocols (like OSPF) handle routing *inside* an Autonomous System, the **Border Gateway Protocol (BGP)** is the essential protocol that connects these systems together.

The Central Challenge

BGP allows a router to determine how to reach a destination located **outside** its own Autonomous System. Without BGP, individual ASs would be isolated "islands" unable to communicate.

2 Routing via CIDR Prefixes

In BGP, we do not route to individual IP addresses. Instead, we route to **CIDRized Prefixes**.

Prefix-Based Routing

- **Notation:** A prefix like `138.16.68/22` represents a subnet or collection of subnets (in this case, 1,024 IP addresses).
- **Forwarding Table:** Entries take the form `(Prefix, Interface)`, mapping a CIDR block to a specific outbound link.

3 Core Responsibilities of BGP

As an inter-AS protocol, BGP provides each router with two critical sets of information:

BGP's Dual Tasks

1. **Obtain Prefix Reachability:** Allows subnets to advertise their presence to the rest of the Internet ("I exist and I am here").
2. **Determine "Best" Routes:** When multiple paths to the same prefix exist, BGP runs a selection procedure based on **policy** and reachability data.

4 BGP Connectivity: eBGP vs. iBGP

BGP operates over semi-permanent **TCP connections on Port 179**. These connections are categorized based on where the routers are located.

Types of BGP Sessions

- **eBGP (external BGP):** A session between routers in **different** Autonomous Systems. These usually connect "Gateway Routers."
- **iBGP (internal BGP):** A session between routers within the **same** Autonomous System. These are used to distribute external reachability info to all internal routers.

5 Advertising and Propagating Routes

Reachability information (the "AS Path") is propagated across the Internet using a combination of eBGP and iBGP.

Step-by-Step Advertising Flow

Suppose AS3 contains prefix **x**.

1. **AS3 → AS2:** Gateway 3a sends an **eBGP** message `"AS3 x"` to Gateway 2c.
2. **Internal to AS2:** Gateway 2c distributes `"AS3 x"` via **iBGP** to all routers inside AS2 (including Gateway 2a).
3. **AS2 → AS1:** Gateway 2a sends an **eBGP** message `"AS2 AS3 x"` to Gateway 1c.
4. **Internal to AS1:** Gateway 1c uses **iBGP** to inform all routers in AS1 of the path `AS2` → `AS3` → `x`.